SINGAPORE SCHOOL

CONTINUAL ASSESSMENT 1

Secondary 1

Computer Science (Part 1)

2026-27

Date: ___ / ___ / ___ **Name: _____**

Instructions for students:

- **Start answering the questions on time.**
- **Follow all the instructions carefully.**
- **Answer all questions.**
- **Revise**

CA 1 Marks (Part – 1)	**25**
CW	**05**
Total Marks Obtained	**30**

1. Choose the correct answer for each of the following multiple-choice questions. (1 x 10 = 10)

1. What is the base of the binary number system?

A) Base 2 B) Base 8

C) Base 10 D) Base 16

Answer: [ ]

2. Convert the 4-bit binary number 1101 into decimal (denary):

A) 11 B) 12

C) 13 D) 14

Answer: [ ]

3. Convert the decimal (denary) number 25 into an 8-bit binary number:

A) 00011001 B) 00010101

C) 00110001 D) 00011100

Answer: [ ]

4. How many binary bits are represented by a single hexadecimal digit?

A) 2 bits B) 4 bits (1 nibble)

C) 8 bits (1 byte) D) 16 bits

Answer: [ ]

5. In the hexadecimal number system, what decimal value does the letter 'E' represent?

A) 12 B) 13

C) 14 D) 15

Answer: [ ]

6. Convert the binary number 1010 1111 into hexadecimal:

A) 9F B) AF

C) B7 D) AE

Answer: [ ]

7. Convert the hexadecimal number 3C into an 8-bit binary number:

A) 0011 1100 B) 0011 1010

C) 0110 1100 D) 0011 1101

Answer: [ ]

8. In binary addition, what is the result of 1 + 1?

A) 2 with carry 0 B) 1 with carry 1

C) 0 with carry 1 D) 0 with carry 0

Answer: [ ]

9. In binary addition, what is the result of adding three ones (1 + 1 + 1)?

A) 1 with carry 1 B) 0 with carry 1

C) 1 with carry 0 D) 3 with carry 0

Answer: [ ]

10. What is the result of adding the 4-bit binary numbers 0101 + 0011?

A) 0111 B) 1000

C) 1001 D) 1010

Answer: [ ]

2. Convert each of the following 8-bit binary numbers into decimal (denary). Show your working. **(1 x 3 = 3)**

(a) 0010 1101

Working & Answer Box:

(b) 0101 0110

Working & Answer Box:

(c) 1001 1011

Working & Answer Box:

3. Convert each of the following decimal (denary) numbers into 8-bit binary. Show your working. **(1 x 3 = 3)**

(a) 42

Working & Answer Box:

(b) 99

Working & Answer Box:

(c) 205

Working & Answer Box:

4. Convert each of the following 8-bit binary numbers into hexadecimal. **(1 x 3 = 3)**

(a) 0100 1101

Working & Answer Box:

(b) 1011 0111

Working & Answer Box:

(c) 1110 1010

Working & Answer Box:

5. Convert each of the following hexadecimal values into 8-bit binary. **(1 x 3 = 3)**

(a) 59

Working & Answer Box:

(b) A6

Working & Answer Box:

(c) F8

Working & Answer Box:

6. Perform binary addition on each of the following pairs of 8-bit binary numbers. **(1.5 x 2 = 3)**
Show all carry bits clearly.

(a) 0010 0111 + 0011 1010

Working & Answer Box:

(b) 0101 1101 + 0010 1110

Working & Answer Box: